

[Secure Exit](#)SIMATS
ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

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CO3-AT3-Debugging and Optimization

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QUESTIONS

Q1ATM Withdrawal – Exception
Handling Debugging
10 marks**Q2**Hospital Registration –
NullPointerException
Debugging
10 marks**Q3**E-Commerce Product Search –
Array Exception Debugging
10 marks**Q4**Bank Account – Race Condition
Debugging
10 marks**Q5**Railway Reservation –
Synchronization Debugging
10 marks

5/5 answered

Q2

Hospital Registration – NullPointerException Debugging

10
marks

A hospital patient registration application crashes when patient information is unavailable.

Debug the following complete Java program.

Task:

1. Identify the runtime error.
2. Modify the program to handle a missing patient name.
3. Handle invalid age input.
4. Use try-catch-finally appropriately.
5. The program must always display "Registration completed".
6. Submit the COMPLETE corrected Java program.

Faulty Program:

```
import java.util.Scanner;

class PatientRegistration {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter patient name: ");
        String inputName = sc.nextLine();
```

```
String name;

if (inputName.equals("NULL")) {
    name = null;
} else {
    name = inputName;
}

System.out.print("Enter age: ");
int age = sc.nextInt();

System.out.println("Patient Name: "
    + name.toUpperCase());

if (age >= 18) {
    System.out.println("Adult patient");
} else {
    System.out.println("Minor patient");
}

System.out.println("Registration completed");
}
}
```

The corrected program must prevent
NullPointerException and handle invalid age input
without terminating unexpectedly.

✳ **Debugging Task:** Find and fix the bugs in the code
shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;
2 import java.util.InputMismatchException;
3 class PatientRegistration
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         System.out.println("enter the patien
9         String inputName= sc.next();
10        String name;
```

```
11     if(inputName.equals("Null"))
12
13     {
14         name = null;
15
16     }
17     else
18     {
19         name = inputName;
20
21     }
22     try
23     {
24
25         System.out.println("enter the age");
26         int age = sc.nextInt();
27         if(name == null)
28         {
29             System.out.println("patient
30         }
31         else
32         {
33
34             System.out.println("patient name:"
35         }
36         if(age>=18)
37         {
38             System.out.println("adult patien
39         }
40         else if(age>=0)
41         {
42             System.out.println("minor patien
43         }
44         else
45         {
46             System.out.println("invalid
47         }
48         System.out.println("registration com
49     }
50     catch(InputMismatchException e)
51     {
52         System.out.println("invalid age"
53     }
54
55
```

56 }

57 }

Test Input

harshitha

1

Output

```
enter the patient name
enter the age
patient name:HARSHITHA
minor patient
registration completed
```

 Pending manual review Submitted